AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I148*

Team Name: *ROBOROOKIES*

Domain: *Intelligent Systems and Digital Futures*

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Problem Statement

Queue-Free Autonomous Printing

Problem Understanding

- Students, Employees, visitors at a high volume institutions.
- At a high volume institution still rely on counter based printing, waiting in queues, staff workload, limited accessibility.
- Users: They have wait in long queue during busy hours, payment delays. Stakeholders: The workload increases during busy working hours, they have fixed time for work and break timimgs.

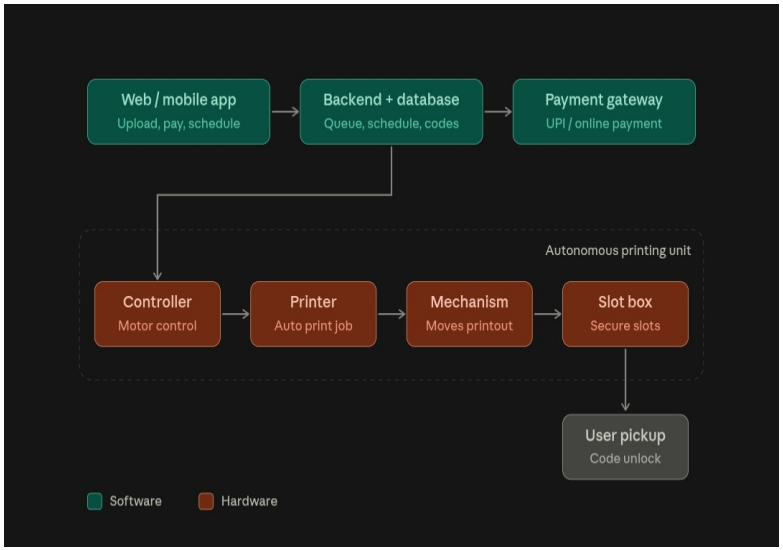
Existing Gaps & Challenges

- Now, we have to wait in a queue and upload the document to their mail. After this we have be patient until the staff takes the printout but we won't receive the document immediately ,before that the payment has to be done either by cash/online.
- User can't tell in advance wether a file will print correctly, so error shows up. Waiting after the mail or messaging the file doesn't help, so user waits anyway.
- 1. Long queue at peak hour. 2. counter hour and staff break. 3. Payment is separate from printing. 4. privacy risk

Our Proposed Solution

- A web based autonomous printing system where students and employees upload, schedule, pay and collects their printout from the assigned slot box , with no queue and no staff involved.
- The user uploads a file through the web, pays online and picks on time. The system automatically places the print in a slot box , and the user can use the one time pin generated to unlock the box.
- No waiting in queues, printing and collection on their own schedule and secure self-service.

Technical Approach



The user uploads a document in the web or mobile app, pays online and picks a pickup time. The backend stores the job, confirms payment and sends it to the controller at the printing unit at the scheduled time. The controller starts the printer and then the mechanism, which places the printout in a free locked slot. The user receives a one-time code, enters it at the slot box and collects the printout without any staff involved.

Tech Stack

Frontend: React web app

Database: PostgreSQL

Backend: Python

AI/ML, APIs, or Hardware: [Add if applicable]

Innovation & Uniqueness

- Esisting print system automate sending files and payment , but user still queues at the booth. This system also automates collection: printouts store in slots and user use the one time pin to unlock, with no staff.
- The delay comes from the queue and staff handling, so the fix is to remove the watching step itself. Scheduled the printing speard demand across the day and slot-box collection work outside counter hours
- One platform that combines upload, online payment, pick-up time scheduling and secur unattended collection.

Feasibility & Scalability

- **Feasibility:** Needed a printer, slot boes, embedded controllers and mechanical module, backend and online payment. Build in 3 steps: web, payment, then the printing unit prototype, then a pilot at one campus printing counter.
- **Scalability:** Add more slots or machinsby registering them in the database. the same platform can serve more locations and more users are handled by scheduling pickups across the day.
- **Risks:** Paper jams or empty torn: sensor alert, operator notifications and automatic refund. Privacy: one time pin for each slot and file deleted after printing. Unclaimed printout are collected for some time limit.

Cost for Product (Optional)

- Development or hardware cost:40,000
- Hosting or maintenance cost: 5,000
- Estimated total: 45,000

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: , student before a deadline: A student uploads an assignment from the hostel, pays online and schedules the print for an hour later. The printout is waiting in a slot when they arrive, and they collect it with a code, skipping the queue.
- Use case 2: employee or visitor outside counter hours: An employee finishing a report after the printing centre closes sends the file from their phone. They collect it from the slot box the same evening, with no staff needed.
- Use case 3: printing centre staff and administration: Staff use the admin dashboard to monitor paper and toner levels, see failed jobs and issue refunds. Their time moves from routine printing to handling exceptions.

Impacts and Benefits

- Benefit to the target users: Students and employees save time because they no longer queue, and they can print and collect on their own schedule, even outside counter hours. Secure self-service also keeps their documents private.
- Social, economic or environmental impact: Socially, it makes printing more accessible and less stressful around deadlines. Economically, staff spend less time on routine jobs, and the centre can serve more users without more manpower. Environmentally, a file check before printing can cut wasted reprints from format errors, and files are deleted after printing.
- Expected measurable outcome: Pickup in under 3 minutes, compared with the 183 seconds in the 2024 prototype. Average user waiting time cut compared with the campus baseline you measure. Jobs completed per day with no staff involved, and the share of jobs that fail.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
9	9.1	Industry, Innovation and Infrastructure	A technology-based printing service that gives users reliable, equal access to printing and replaces manual counter work with automation.
4	4.1	Quality Education	[Better campus facilities for students, who spend less time in queues and can print assignments and study material at any hour.

Individual Member Contribution

Member	Contribution or responsibility
Trishul Sankar. G	Web and mobile app, backend and database: upload, scheduling and job tracking
Jeyarjun R	Online payment, one-time code or QR security, and file privacy (auto-deletion)
[Subradheepan K	Embedded controller, sensors and printer integration
Shashwat Harith	Mechanical slot box and mechanism, plus research and presentation

Yukti ID

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